

Lesson 8: Two Qubits and Entanglement

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Hadamard, CNOT, Bell-state intuition, and correlated measurements

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Learning Outcomes

- Explain what two qubits are.
- Understand the CNOT gate at an intuitive level.
- Describe entanglement as a correlation created by a quantum circuit.
- Interpret the observed 00 and 11 counts.

From One Qubit to Two Qubits

- Two independent classical bits have four combinations: 00, 01, 10, 11.
- Two qubits also have four computational basis states.
- Quantum circuits can create correlations that cannot be described as two independent classical coin flips.

The Bell-State Construction

$$|00\rangle \xrightarrow{H \otimes I} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{CX} \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

- Hadamard acts on qubit 0.
- CNOT uses qubit 0 as control and qubit 1 as target.
- The resulting state is an entangled Bell state.

Read the Two-Qubit Code

```
qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
qc.measure_all()
simulator = Aer.get_backend('aer_simulator')
job = simulator.run(qc, shots=1000)
result = job.result()
counts = result.get_counts(qc)
print(counts)
```

- `QuantumCircuit(2)` creates two qubits.
- `h(0)` creates superposition in the first qubit.
- `cx(0,1)` correlates the qubits.
- `measure_all()` measures both qubits.

Interpreting the Output

{00: 509, 11: 491}

- The notebook records only 00 and 11 in the displayed run.
- The counts are close to 50–50.
- 01 and 10 do not appear in the recorded output.
- That correlation is the observable signature of the entangled state in this simple experiment.

Why CNOT Is Important

- The control qubit determines whether an X operation is applied to the target.
- Starting from the superposition created by H, CNOT links the two qubits.
- Entanglement becomes a building block for many quantum algorithms and protocols.

Beginner Activity

- Remove the CNOT and rerun.
- Predict whether 01 and 10 will now appear.
- Restore CNOT and compare the distributions.

- Two qubits have four computational basis states.
- H creates superposition; CNOT creates correlation.
- The Bell-state experiment yields correlated 00/11 measurements.

Entanglement is correlation created by quantum evolution, not ordinary data duplication.

Quick Check

- ❶ What is the control qubit?
- ❷ What is the target qubit?
- ❸ Why do 01 and 10 disappear in the recorded output?